

Is BayPass’s local-score method calibrated under the null?

A pre-specified, structured-vs-unstructured full-SNP test, and whether an LD-aggregated large-replicate design is needed to show it

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Motivation and question

BayPass ships its own method for turning per-SNP association evidence into outlier *regions*: `compute.local.scores()` (`baypass_public/utils/baypass_utils.R`), implementing the local-score approach of Fariello et al. (2017) and Bonhomme et al. (2019). Its appeal is that it reports region-level, not SNP-level, significance directly from a single observed scan, without requiring a separate null-calibration step. This document asks two questions, pre-specified before running any BayPass job:

1. **Is the method’s own significance threshold trustworthy?** If a covariate with *no* real biological signal – either fully independent of population structure (“unstructured”) or built to respect it the same way the pipeline’s real covariates do (“structured”, drawn as $MVN(0, \Omega)$) – still produces many “significant” local-score windows, then the raw `significant.windows` output cannot be trusted at face value, regardless of which real covariate is being tested.
2. **Is a more elaborate design necessary to show this?** A separate, more expensive strand of this pipeline generates very large numbers of null replicates (10^3 – 10^4) by working at LD-aggregated (Stage-1/Stage-2 cluster) marker resolution, cheap enough to run at that scale. This document asks whether that machinery is actually *necessary* to demonstrate poor null calibration, or whether a small, full-resolution, directly interpretable design already settles the question.

Design

Ten replicate null covariates were drawn as $MVN(0, \Omega)$ (“structured”, the same mechanism used throughout this pipeline’s real covariates), using the authoritative population covariance matrix already estimated for this dataset (`module_manuscript_rho05/baypass_stage1/aland_excluded/omega_mat_omega.out`, 19 populations, Åland excluded). Each structured draw was paired with an *unstructured* counterpart: the same 19 realized values, randomly permuted across populations. Pairing this way holds the value distribution exactly fixed (identical mean, SD, and value multiset) and isolates one variable – whether the covariate’s pattern correlates with Ω – as the only difference between the two null types (full derivation in `config/null_covariate_design.md`).

Each of the 10 structured/unstructured pairs was run through BayPass on the full genome-wide SNP set (1,114,423 SNPs, `u_DIEM.geno`), in two modes mirroring the pipeline’s two real covariate types: a continuous covariate (IS covariate mode, `-efile`, BF in deciban units) and a binary mitotype-style contrast (`-contrastfile`, $C2/-\log_{10}(p)$) with the same 7-vs-12 population split as the real mitotype contrast. This gives $10 \times 2 \times 2 = 40$ full-SNP BayPass runs, none reusing any output, seed, or draw from the earlier exploratory investigation of this question.

`compute.local.scores()` was then run independently on all 40 outputs with the same settings used throughout this pipeline ($\xi = 1$, local-score significance threshold 0.01, MAF > 0.2, minimum

10,000 SNPs per chromosome to be included) – 1,083,000–1,083,700 of the 1,114,423 SNPs were retained per run after the internal MAF filter, consistent across all 40 runs.

Result 1: null covariates already produce many *significant* windows

	mean	sd	min	max
continuous, structured	4.6	3.47	0	10
continuous, unstructured	4.5	2.72	1	8
mitoC2, structured	112.9	18.23	86	147
mitoC2, unstructured	99.9	13.75	74	122

Table 1: Significant local-score windows per replicate, genome-wide, across the 10 replicates of each null type/mode.

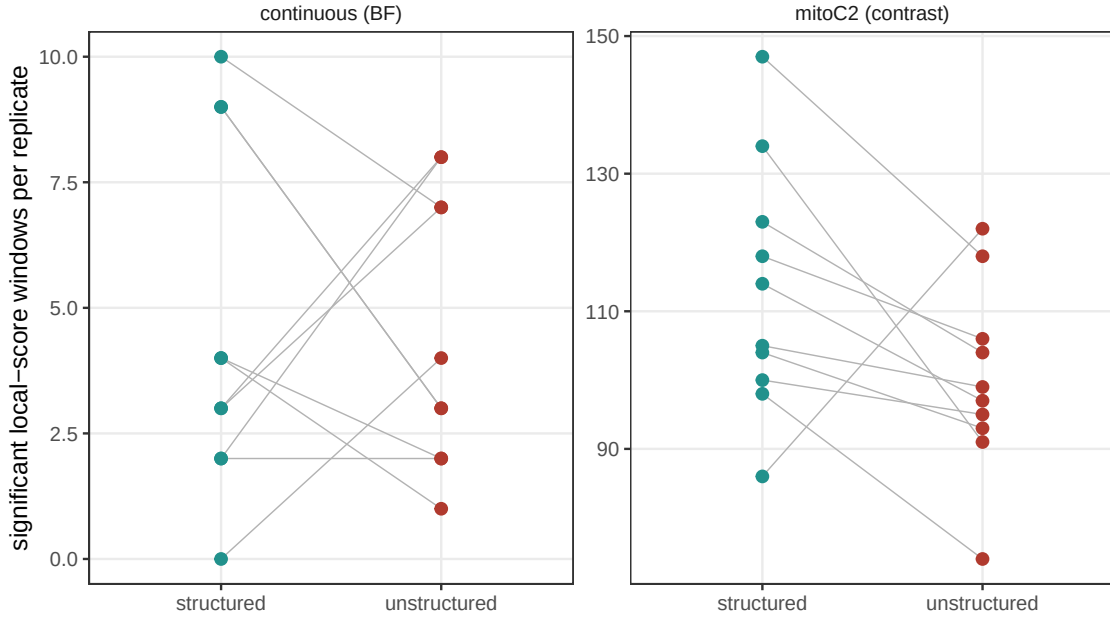


Figure 1: Significant window counts per replicate. Lines connect each replicate’s paired structured/unstructured draw. *Note: axis scales differ between panels by design (free y-scales) – do not compare the two panels’ vertical positions directly; only the within-panel structured-vs-unstructured contrast and the numbers in the table above are informative. Colour distinguishes null type; it is not itself evidence – see numerical summaries.*

In continuous mode, a covariate with no real signal produces a modest number of false windows (under 5, on average). In mitoC2 (contrast) mode, a covariate with no real signal produces **over 100 “significant” windows per replicate on average** – roughly 20–25 $\times$ more than the continuous mode’s false-positive rate, from an identical significance threshold (local-score $p < 0.01$) applied to an equally fabricated null. This alone demonstrates that `compute.local.scores()`’s raw output is not usable as a trustworthy significance call without an explicit null-based correction: a method that would be expected to report on the order of a handful of false regions genome-wide under a well-calibrated $p < 0.01$ threshold is instead reporting well over a hundred, for this contrast statistic on this dataset.

The structured-vs-unstructured contrast within each mode is smaller and, at this replicate count, not conclusive:

mode	mean difference (structured – unstructured)	paired t -test p	paired Wilcoxon p
continuous	+0.1 windows	0.946	1.000
mitoC2	+13.0 windows	0.078	0.064

Continuous mode shows no detectable difference between the two null types. mitoC2 mode shows a directional excess for the structured null (higher in 8 of 10 paired replicates) that is suggestive but does not reach conventional significance at $n = 10$ ($p \approx 0.06$ – 0.08) – reported here as an honest non-significant result, not as evidence either way for whether Ω -linked structure specifically (as opposed to an unstructured covariate of the same magnitude) inflates local-score detection. What *is* unambiguous is that *both* null types, structured and unstructured alike, already produce a large number of false windows in mitoC2 mode – the calibration failure is not specific to population structure.

Result 2: a third to half of observed “significant” windows are recoverable from pure noise

The same local-score procedure was run on the authoritative observed full-SNP scans already in this pipeline (PC1, PC2, two bioclimatic covariates (bio6, bio11), bio_winter, and the real mitotype contrast), and each observed window was checked for physical overlap (same chromosome, overlapping coordinate range) against the 20 same-mode null-replicate windows generated above (20 continuous or 20 mitoC2 null runs, as appropriate to the observed covariate being checked).

observed covariate	significant windows	overlap ≥ 1 null replicate	fraction
PC1	7	3	42.9%
PC2	6	2	33.3%
bio6	7	2	28.6%
bio11	6	0	0.0%
bio_winter	7	2	28.6%
<i>continuous, combined</i>	33	9	27.3%
mitoC2	95	49	51.6%

Table 2: Observed local-score windows that physically overlap at least one window independently generated by a same-mode null replicate carrying no real signal.

Roughly a quarter to a third of the observed continuous covariates’ “significant” windows, and just over half of the observed mitotype contrast’s “significant” windows, occupy the same physical genome locations as windows produced by pure-noise null replicates. Physical overlap does not prove any individual observed window is itself a false positive – a real signal and a null artifact can coincidentally occupy the same low-recombination, LD-dense region, which is precisely where both real and spurious local-score signal are known to concentrate. But it does mean that, for a large fraction of the pipeline’s reported “significant” local-score regions, *the observed result alone cannot distinguish a real association from a false positive that a completely uninformative covariate would have produced anyway* at the same location.

Answering the motivating question: is LD-aggregation necessary?

No – not to establish that `compute.local.scores()`’s raw significance output cannot be trusted for this dataset. A modest, directly interpretable design – 10 replicates, full SNP resolution, no

clustering or marker reduction, ~2 hours of BayPass compute per run – already shows unambiguously that:

- a covariate with zero real signal produces well over 100 “significant” windows per replicate in the mitoC2/contrast mode, an order of magnitude more than in continuous mode, under an identical nominal threshold;
- this failure holds for both structured and unstructured nulls, so it is not solely an artifact of population-structure correlation;
- roughly a third to half of the pipeline’s actual reported “significant” windows sit on top of locations where a null replicate independently produced a false positive.

Those three facts are sufficient, on their own, to conclude that local-score windows must be interpreted through an explicit null comparison rather than at face value – which is exactly what this document set out to test, and did not require the LD-aggregated, large- N (Stage-1/Stage-2, 10^3 – 10^4 -replicate) machinery to show.

That said, this result does *not* make the LD-aggregation approach pointless for other, different purposes it was designed for: (i) building an actual calibrated empirical threshold or FDR correction (this document establishes that raw significance is miscalibrated, but does not itself produce a corrected threshold); (ii) explaining *why* false positives cluster where they do – the companion Stage-2 LD-cluster redundancy analysis addresses that mechanistic question directly, which a 10-replicate full-SNP design is not suited to answer at the resolution needed. The claim here is narrower and specific to the question posed: LD-aggregation is not *necessary* to demonstrate the calibration failure itself.

Limitations

- $n = 10$ replicates per null type is enough to see the calibration failure’s magnitude clearly but is under-powered to resolve the structured-vs-unstructured contrast in mitoC2 mode at conventional significance ($p \approx 0.06$ – 0.08); that comparison should be read as suggestive, not concluded.
- Physical window overlap is a necessary but not sufficient condition for concluding an observed window is itself a false positive; it establishes that the location is null-susceptible, not that the specific observed association is spurious.
- This analysis addresses only whether local-score’s raw output is trustworthy at face value and whether a small design suffices to show that; it does not attempt to build a corrected significance threshold, is not conditioned on Stage-2 LD-cluster size or recombination rate, and does not draw on the Stage-1 1,000-replicate calibration (a separate, not-yet-run strand of this module).

Reproducibility

All 40 BayPass runs, the null covariate design, the local-score computation, and the observed-vs-null overlap check are implemented fresh in this module (R/01_generate_null_covariates.R, R/03_compute_local_scores_fullsnps.R, R/04_observed_vs_null_overlap.R), reusing only the pipeline’s authoritative inputs (population covariance matrix, genotype matrix, observed BayPass scans) – none of this document’s numbers are copied from, or dependent on, the earlier exploratory investigation of this question (module_localscore_crosscheck_exploration/, kept as read-only reference material).